Using XBRL to conduct a large-scale study of discrepancies between the accounting numbers in Compustat and SEC 10-K filings

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January 30, 2014

Abstract

Compustat accounting database is frequently used for both research and decision-making. It has been documented that information found in Compustat database differs from both the information found in other accounting databases and the information disclosed in corporate financial filings (San Miguel 1977; Rosenberg and Houglet 1974; Yang et al. 2003; Tallapally et al. 2011; 2012; Boritz and No 2013). In this study, we conduct the first large-scale comparison of Compustat and 10-K data. Specifically, we compare 30 accounting line items of approximately 5,000 companies for the period from October 1, 2011 to September 30, 2012. We find that the values reported in Compustat significantly differ from the values reported in 10-K filings. We also find that the amount and magnitude of the original data alterations introduced by Compustat depend on the type of the accounting item and company characteristics such as industry and size.

Keywords: Compustat, 10-K, XBRL, data quality, data discrepancies, data aggregators

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The authors thank Michael Alles, Khrystyna Bochkay, Valentin Dimitrov, Miklos Vasarhelyi, participants of 2013 AAA Annual Meeting, and seminar participants at Rutgers University for their helpful comments and suggestions.

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1 Introduction

Compustat is a popular source of financial information for both academics and practitioners. It has been maintained by Standard and Poor's[1] company since 1962. It has been used extensively in literature to support various findings and hypotheses. In fact, many accounting empirical papers that study U.S. firms make use of Compustat data. However, it has been questioned how reliable the data found in Compustat is (and by extension in other accounting databases). Prior studies have shown that Compustat data may differ from the real-world financial data (San Miguel 1977; Kinney and Swanson 1993; Tallapally et al. 2011; 2012; Boritz and No 2013) and data found in other accounting databases (Rosenberg and Houglet 1974; Yang et al. 2003).

Compustat uses companies' source reporting documents (such as 10-Qs and 10-Ks) to populate its fundamentals data set. However, the original data from the source documents is standardized per Compustat's variable definitions for "ensuring. . . comparable data across companies, industries and time periods without reporting biases or data discrepancies" (Capital IQ Compustat 2013). There are two important questions related to this fact: should Compustat standardize the original data reported by companies, and what is the extent of this standardization. In this paper we address the latter question, although we also find some empirical evidence suggesting that data standardization may, in fact, decrease company data comparability in some cases. In addition to data standardization, Compustat numbers may differ from the original numbers reported by companies due to typos, missing values, not-up-to date values, etc. It is important to assess how much difference there is between Compustat data and the original data, since this difference may affect the results of accounting studies that utilize Compustat.

The only way to assess the extent of data alteration in Compustat is to compare Compustat numbers to the numbers in original reports. Unfortunately, this is a very costly procedure, if done manually. Prior studies that contrasted Compustat and original reports'

1. Standard & Poor's is a division of The McGraw-Hill Companies.

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numbers had an important limitation of using small samples in their analyses. Small samples may not fully represent the population, and may omit many non-common observations (e.g., discrepancies). In this paper, we utilize the power of the XBRL reporting technology to automatically extract accounting numbers from XBRL financial reports and compare them to Compustat numbers. The SEC has mandated the use of XBRL reporting by all U.S. GAAP filers starting from June 2011. The methodology that we develop in this paper allows us to do a large-scale comparison of Compustat and original reports' numbers. To our best knowledge, this is the first study to conduct such a large-scale data comparison.

Specifically, we study the amount and magnitude of discrepancies between Compustat and 10-K reports for the period from October 1, 2011 to September 30, 2012.[2] Our analysis includes more than 5,000 companies, and 30 accounting items that are often used in accounting empirical research. We limit our analysis to 10-K reports only since they are audited, and to one 10-K report per company to make our sample unbiased and study whether company haracteristics are related to the amount and magnitude of discrepancies.

We find that Compustat significantly alters numbers reported in the 10-K filings. In particular, we find that Compustat values of 22 out of 30 analyzed variables significantly differ from values reported in the 10-K filings. We also find that the type of statement where variable is reported and company characteristics such as industry and size are related to the amount and magnitude of discrepancies.

In addition to our findings, we develop a methodology for an automated large-scale comparison of Compustat and XBRL data. The methodology comprises of several steps: data extraction, data merging, concept mapping, difference calculation, automated error detection, and discrepancy analysis. The methodology will not only be useful to study the extent of discrepancies between the two data sets, but also to validate financial data used for accounting empirical research.

2. Although we use the XBRL technology to extract data from XBRL 10-K reports, our objective is not to compare Compustat and XBRL 10-Ks, but rather to compare Compustat and plain-text 10-Ks. XBRL 10-K reports sometimes contain errors that make them different from plain-text 10-Ks. In this study, we try to filter out all XBRL-related errors using both automated and manual procedures.

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The paper is organized as follows. Section 2 reviews academic literature relevant to this study. Section 3 discusses types of data alterations in Compustat. Section 4 presents Compustat data cross-verification methodology. Section 5 reports the main findings of this study. Finally, Section 6 concludes the paper.

2 Relevant literature

In this section, we review relevant literature that studies the impact of discrepancies in accounting databases. Since there are two main causes of discrepancies – errors and different data definitions – we consider the respective literature separately.

2.1 Erroneous data

Rosenberg and Houglet (1974) is the first paper to consider the quality of data found in Compustat. The authors match and merge Compustat and CRSP data on monthly price relatives for 844 industrial sector companies from January 1963 to June 1968, and for 97 utilities sector companies from March 1962 to June 1968. In total, they compare 41,296 monthly price relatives between Compustat and CRSP data sets. Out of these, they find 1,202 (2.91%) to be erroneous including 294 (0.71%) that differ by more than 5%, and 125 (0.3%) that differ by more than 20%. The authors compare 34 discrepancies between utilities price relatives to the original published sources; 4 (12%) discrepancies were due to CRSP errors, and 30 (88%) were due to Compustat errors suggesting that CRSP is a more reliable database for monthly price relatives. The study finds these errors to change the data distribution, with higher moments being affected more significantly.

Bennin (1980) did a follow-up study of Rosenberg and Houglet (1974) using updated Compustat and CRSP data, and for a longer time period. He compares monthly relative prices of both industrials and utilities for the period from January 1962 through July 1978. Out of 187,460 prices, he finds 471 (0.25%) to differ by more than 5%. This is a huge drop

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of error rate compared to Rosenberg and Houglet (1974) – 0.25% versus 0.71%. The study suggests that Compustat had tried to correct errors due to Rosenberg and Houglet (1974) study.

It should be noted that both Rosenberg and Houglet (1974) and Bennin (1980) consider only a small piece of information available in Compustat - monthly prices. These are usually straightforward to record and do not require any preprocessing or adjustments. For example, consider Cost of Goods Sold (COGS). Compustat may adjust COGS reported the in 10-Ks by subtracting Depreciation if it was included in company's 10-K (e.g., Amazon.com's 2011 10-K)[3]. This increases the probability of error since both the original COGS and Depreciation should be recorded properly. In addition, the adjustment process should be carried out correctly. Also, the error rates found in Compustat, although small, may significantly change the inferences drawn from the data. This is supported by Beedles and Simkowitz (1978). The authors replicate the study of McEnally (1974) that investigates the return behavior of high-risk common stocks extracted from the CRSP database. After correcting the CRSP database errors, Beedles and Simkowitz find that the results changed significantly due to changes in higher moments of data distribution. Such changes could be even more significant if Compustat data was used instead of CRSP since it was found to be more erroneous.

San Miguel (1977) is the first study to compare Compustat data to the 10-K reports. Specifically, he compares Research and Development (R&D) expense data in Compustat with the original 10-K reports for a sample of 256 companies that reported R&D expense in 1972.[4] He finds 78 (30%) discrepancies between the two datasets. At least 52 (66.67%) discrepancies were due to errors in Compustat. Out of those 15% were rather significant - the discrepancies amounted to more than 100% of the Net Income values of the respective companies. The author also finds that there was insufficient information in the 10-K reports

3. According to Compustat's manual, Compustat definition of COGS does not include Depreciation. There is a special data footnote "BD" in Compustat that indicates that the amount was reduced by the value of Depreciation.

4. In 1972, 1,357 Compustat firms reported R&D expense. A sample used in San Miguel (1977) represents 19% of those.

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to classify the remaining 26 discrepancies. He attributes part of the errors to the new rules for the 10-K Form R&D disclosure requirements that became effective in 1972, and to the complexity of some 10-K reports that Compustat personnel had to deal with. However, he still concludes that Compustat data is not very reliable.

Kinney and Swanson (1993) examine the accuracy of the tax data in Compustat. The authors randomly select 100 Compustat companies from fiscal 1985 listings. They compare 19 tax variables as reported by Compustat and the original financial reports for the years of 1986-1988. The error rates for the considered variables ranged from 0.76% to 11.65% with the error magnitudes being substantial). The authors find the tax error rates to be: 1) high for utility companies, 2) low for balance sheet items, and 3) high for cash flow statement items. They also find a significant number of missing values for some variables whereas these values are reported in the financial items.

Kern and Morris (1994) study the data differences between Compustat and Value Line databases. The authors compare Sales and Total Assets data for the years of 1971-1990. They find that there is no statistically significant difference with respect to Total Assets. However, they find significant differences between Sales values. The number of discrepancies between Sales ranged from 31.4% to 33.9% across different years. The years of 1985-1990 had the highest number of material discrepancies (137 in Total Assets and 378 in Sales). The authors examined the annual reports for those years to identify the sources of the discrepancies. Many discrepancies (40.9% for Total Assets and 15.1% for Sales) were due to Value Line reporting data in currency other than the U.S. dollars. The largest source of discrepancies for Sales was the difference in data definitions used in the databases. For example, Value Line include only the income items in Sales that are related to the company's major line of business, while Compustat reports Sales for the entire consolidated entity; this resulted in 47.6% of all discrepancies. But the authors also find some unexplainable differences (at least 1.5% of all discrepancies).

Yang et al. (2003) compare Compustat and Value Line values of 7 frequently used account-

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ing variables for the period of 1976-1981.[5] Out of 10,353 observations (of 1,479 companies), 1,284 (12.5%) were discrepancies larger than 1%. To identify the cause of discrepancies, the authors draw a subsample of 200 companies and compare the Compustat and Value Line 1981 data to the original 1981 financial statements data. Out of 1,400 observations, they find 320 mismatches. Out of those mismatches, 185 (57.81%) were explainable discrepancies (i.e., discrepancies due to different data definitions, currency and industry factors), and 135 were unexplainable that were either due to errors or undisclosed coding rules. It is interesting to note that Compustat amounted for more unexplained discrepancies – 99 (73% of all unexplained differences) – than Value Line. The authors also replicated the part of the Rosenberg and Houglet (1974) study by computing and comparing data distribution moments of Compustat and Value Line databases. At least for some variables (e.g., Current Assets) the differences were found to be statistically significant. The authors conclude that accounting databases of data aggregators contain a certain level of erroneous data and do not always agree on data definitions and industry classifications, and that users of accounting databases must be aware of those differences.

Tallapally et al. (2011) compares EDGAR Online[6] and Compustat values of the Cost of Goods Sold item for a subset of DOW 30 companies for the fiscal year of 2009. Out of 26 companies considered, there was only one match in COGS between Compustat and EDGAR Online. The average magnitude of remaining 25 discrepancies is 14.23% with Compustat reporting, overall, lower numbers. The authors were not able to reconcile the numbers, but assume that these differences are due to data definitions used in Compustat.

Tallapally et al. (2012) is a very similar study to the previous one with the same authors (Tallapally et al. 2011). It compares Compustat and 10-K XBRL (eXtensible Business

5. Yang et al. (2003) examined The Accounting Review, The Journal of Accounting Research and The Journal of Accounting and Economics for the 1976-1981 period to find the most frequently used variables in accounting research. They identified the following variables: Total Assets, Net Sales, Inventories, Net Income, Current Liabilities, Depreciation, Depletion, and Amortization, and Gross Plant.

6. Not to be confused with SEC's EDGAR. EDGAR Online is a public for-profit company that provides SEC fillings information extracted from SEC's EDGAR.

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Reporting Language) data for the same year (2009) and for almost the same set of companies[7] but with regards to Sales/Revenue item instead of COGS. The data extracted from 10-K XBRL should be similar (most likely the same) to the data extracted from EDGAR Online.[8] The authors find differences between Compustat and 10-K XBRL filings with respect to Sales/Revenues, but not as many and not as large as in the study that compared COGS values. Namely, the authors find 6 discrepancies (22% of all observations) with the average magnitude equal to 5.19%. As in the previous study, the authors do not reconcile the discrepancies.

Boritz and No (2013) manually compare financial items reported in 150 XBRL 10-K filings of 75 companies to the corresponding items provided by three data aggregators: Compustat, Yahoo Finance, and Google Finance. The study finds that around 50% of financial items that are reported in XBRL 10-Ks are not present in the aggregators' data sets, and that the percentage of mismatches between XBRL and aggregators' data ranges from 4.8% to 8% with 56% of all differences being material. Out of three financial statements – Balance Sheet, Income Statement, and Cash Flow Statement – Balance Sheet was associated with the smallest percentage of mismatches. In addition, Compustat data resulted in the largest percentage of mismatches (44.3%) and the lowest number of omissions (50.9%).

The most related studies to ours are Tallapally et al. (2011); Tallapally et al. (2012), and Boritz and No (2013). Therefore, we would like point out some key differences with them. Firstly, this study focuses on the accounting data provided by Compustat. The objective is to compare Compustat numbers to the numbers found in the 10-K filings. Although, we utilize XBRL 10-K filings to extract the numbers, we do not aim to compare Compustat and XBRL data since XBRL data also may differ from plain-text 10-K data. In our study, we try to remove all XBRL 10-K values that differ from 10-K values (using both manual and automated procedures described in §4). Secondly, we conduct a large-scale comparison

7. In this study (2012), the authors added additional company to their sample, making total number of companies equal to 27.

8. EDGAR Online extracts information directly from SEC's XBRL filings.

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that involves more than 5,000 companies (as opposed to 75 in Boritz and No (2013) and 27 in Tallapally et al. (2012)) by automating the comparison procedure. Thirdly, we study not only the amount of discrepancies between the data sets, but also the magnitude of discrepancies and their effect on non-discrepancy observations. Fourthly, we develop and present a methodology for an automated (as opposed to manual) XBRL and Compustat data comparison. This methodology is critical for our study since such a large-scale study would not be possible without it. Finally, some of our findings differ from the above mentioned studies (e.g., we do not find significant differences in Total Liabilities between Compustat and 10-K data as Boritz and No (2013) do).

2.2 Data definitions and comparability

Stone (1968) is one of the first studies to caution researchers about the use of the digital accounting databases and Compustat in particular. The study recognizes several shortcomings of Compustat in terms of data comparability. First, it argues, the Compustat annual data inherits SEC 10-K filings weaknesses. Companies may not be directly comparable based only on the 10-K reports since they may use different accounting, e.g., FIFO versus LIFO, or straight line depreciation versus accelerated depreciation, etc. Second, the data definitions used to adjust 10-K data to make it more comparable may fail because of the previous point and may significantly alter the accounting used by a company. Moreover, it creates the appearance of uniformity of data, and a user of Compustat data may draw wrong conclusions assuming items to be comparable. Also, users not fully aware of data definitions are likely to run into problems when creating they own financial ratios or proxies. Last but not least, the study argues that Compustat does not provide qualitative disclosures that may be essential to understand the financial position of a company.

Thies and Revsine (1977) examine the implications of Compustat definitions and policies for the Capital Expenditures item on the empirical inflation accounting research, and specifically for the purpose of asset layering. Empirical inflation accounting studies often

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require to transform conventional accounting numbers to their inflation-adjusted estimates. This is generally done by arraying fixed assets in layers with respect to their acquisition (cost and dates). The study argues that Compustat item Capital Expenditures cannot be used effectively for the purpose of fixed asset layering since: 1) Compustat definition of the item does not include fixed assets acquired through merger or acquisition, and 2) Compustat report capital expenditures net of retirements (if this net figure is shown in a financial report). To test the adequacy of Compustat Capital Expenditures data, the authors impute fixed assets retirements from the Compustat data for the years 1960 through 1974 for S&P 425 industrial firms. They use a simple criterion: check how many imputed retirements are negative.[9]. The proportion of negative values ranged from 7.5% to 30.7% across different years, indicating that Compustat data is too distorted to be used for the purpose of fixed asset layering. The authors suggest to use SEC's 10-K data directly instead of Compustat data, even though it is less accessible and convenient.

Collins and O'Connor (1978) criticize Eskew (1975) study due to its failure to adjust data across different databases: Compustat, Moody's Industrial Reports, and companies' annual reports. They argue that data definitions in Compustat differ from the ones in Moody's and annual reports. The researchers should take special care of data definitions when matching and merging data from several databases, since it can lead to false conclusions. Collins and O'Connor replicate Eskew (1975) study by addressing shortcomings of the latter. The results obtained differ significantly from the original ones.

Guenther and Rosman (1994) study the differences between SIC codes assigned to companies by Compustat and CRSP databases and their effect on accounting research. Using a sample of 1,810 companies, the authors find 1989 Compustat and CRSP databases disagree: on 71% of all companies at the 4-digit SIC level, on 54% of companies at the 3-digit level, on 38% of companies at the 2-digit level, and on 22% of companies at the 1-digit level. The authors also examine the homogeneity of Compustat and CRSP. They find that the Com-

9. Clearly, fixed assets retirements cannot be negative

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pustat SIC code classification yields both higher correlation of intra-industry monthly stock returns and lower variances of intra-industry financial ratios than CRSP SIC classification. To test the impact of the differences in SIC classification between two databases, the authors replicate the study of Freeman and Tse (1992) using both Compustat and CRSP SIC codes. The results obtained using Compustat codes are similar to the results of Freeman and Tse since they also use Compustat. However, the results obtained by utilizing CRSP codes were significantly different from the original ones.

Kahle and Walkling (1996) do a follow-up study of Guenther and Rosman (1994) by examining the differences in SIC classification of Compustat and CRSP databases for approximately 10,000 firms over the years of 1974-1993. They find similar level of discrepancies as Guenther and Rosman: there is 79% disagreement at the 4-digit SIC level, and 35% disagreement at the 2-digit SIC level. To further analyze the impact of these differences, the authors choose six financial characteristics[10], draw a random sample of firms for each characteristic, and conduct simulations to measure the power and specification of Compustat and CRSP SIC classifications on based on these random samples. The authors find that Compustat SIC classification is better at detecting abnormal performance[11] than the CRSP SIC classification.

3 Data alterations in Compustat

As noted above, Compustat data items may differ from the original accounting data items disclosed by companies in their financial reports. We identify four reasons for having such differences:

1. Compustat transformed original value to match Compustat's standard definition of the variable,

10. Specifically, the authors choose: Operating Return on Sales, Operating Return on Assets, Leverage, Asset Turnover, Payout ratio, and Market-to-Book ratio.

11. Abnormal performance in Kahle and Walkling (1996) is simulated by adding errors to the variables to increase the difference between sample and control firms' characteristics.

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2. Compustat's value is erroneous (due to typos, rounding, etc.),

3. Compustat's value is not up to date,

4. Compustat does not provide a value for the data item (i.e., missing data).

Compustat does not explicitly indicate that an alteration has been made to a number reported in a company's financial statement, except in rare cases.[12] This presents a problem since a user of the Compustat data set may be unaware of a change that has been made to the original number. Moreover, the original number is never reported in Compustat. Hence, there is no straightforward way to estimate the amount of difference between the original value reported in 10-K and standardized value reported in Compustat. In addition, as noted in Stone (1968) it is not clear whether Compustat should alter the original data in the first place. For example, Compustat reports Total Assets of American Water Works Company at the end of December 2011 to be \$13,809,643,000, while the value of Total Assets in the 10-K is \$14,776,391,000. The difference is the value of Contributions in Aid of Construction of \$966,748,000.[13] By not including this item in the Total Assets, Compustat underreports the amount of resources that the company has. We believe that this adjustment does not enhance company comparability, and that it actually reduces comparability. Unfortunately, accounting researchers rarely question Compustat's data alterations and take them as given.

Erroneous data is another reason why Compustat numbers may differ from the original numbers reported by companies. Unlike standardized data, errors are unintentional alterations of the original or standardized data. Errors may occur due to input typos, use of wrong accounts and balances, data misinterpretation, data rounding, data calculations, etc. The danger of having errors in the data is that they are hard to identify – they may appear

12. A very small number of alterations is reported through footnotes. For example, footnote "JE" indicates that the reported value differs form the reported amount by deferred taxes.

13. American Water Works Company in its 10-K describes Contributions in Aid Of Constructions as follows: "Regulated utility subsidiaries may receive advances and contributions from customers, home builders and real estate developers to fund construction necessary to extend service to new areas . . . Advances that are no longer refundable are reclassified to contributions in aid of construction. Contributions in aid of construction are permanent collections of plant assets or cash for a particular construction project."

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as standardized numbers. For example, Compustat reports the Gross Profit of Transwitch Corp for the year of 2011 to be $19,932,000, while the company reports the value of Gross Profit to be $17,932,000. The difference of 2 million U.S. dollars might be due to Compustat's adjustment, although we failed to reconcile this discrepancy. It may also be due to input typo since the two numbers differ only in one digit.

Accounting researchers are well-aware of the existence of erroneous data in popular data sets. Sometimes they assume that errors in the data will result in some extreme values. Therefore, they use outlier detection techniques to find extreme values, and either delete them or transform them (using techniques such as winsorising). However, this approach is not ideal since erroneous values do not have to be extreme, and extreme values do not have to be erroneous. Extreme values that are not erroneous should be included in a study sample since they represent possible states of the object studied. Moreover, deleting extreme observations is very likely to affect the output of most empirical models due to their sensitivity to outliers, and this may drastically change the results of the study.

A different type of data discrepancy between Compustat and companies' financial reports is when Compustat does not update accounting numbers due to restatements. Standard & Poor offers products that provide historical time snapshots of financial data (e.g., Compustat Point-In-Time). However, Compustat's fundamental annual data should be updated on a monthly basis.[14] Unfortunately, in some cases, Compustat fails to update its data items in a timely manner that results in inaccurate data. For instance, on March 15, 2012, ADA-ES reported its Net Loss to be $19,851,000; however, on October 19, 2012, the company has restated its Net Loss to be $22,819,000. As of February 22, 2013, Compustat has not updated the originally reported number with the restated one. The resulting difference of $2,968,000 is clearly material since it is more than 10% of the Net Loss.

Data that are not updated should be considered separately from erroneous data since there were no errors in recording the original (not restated) numbers from financial reports.

14. According to Compustat's Manual, data is updated as frequently as daily.

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On the other hand, the fact the numbers were restated indicates that there is a material difference between the original and updated numbers. Hence, using accounting data that has not been updated may result in significantly different results in an empirical study.

Another problem with the Compustat data set is the absence of data for some variables. For example, Compustat did not report the Total Assets of Airwave Labs ($1,139,182) at the end of 2012, although Compustat did report the company's Total Liabilities and Stockholder's Equity. Accounting researchers usually drop observations with missing data, or in rare cases substitute missing data with estimated values (e.g., industry averages). Clearly, in some cases a variable is not applicable to a company (e.g., Sales), or may not be reported in financial reports (e.g., non-GAAP measures). However, if an accounting number is reported in a financial statement, and there is a corresponding Compustat variable for that number, then it should be present in the Compustat's data set. Missing data does not allow researchers to study the whole population, and similarly to data alterations, may change the results of empirical studies.

4 Cross-verification of Compustat annual data with 10-K numbers using XBRL

The best way to assess the data validity in Compustat is to compare the data reported in Compustat to the original data reported in companies' financial reports. In particular, the annual accounting numbers in Compustat can be cross-verified with the accounting numbers found in the 10-K financial reports.

However, a large-scale comparison of accounting data items reported in Compustat and corporate 10-K filings would be a very costly procedure, if done manually. It was the only option not so long ago. However, since the SEC mandated the use of the XBRL reporting technology in the SEC filings for all U.S. GAAP filers for the fiscal years ending on or after June 15th, 2011, it has become possible to automate this process.

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4.1 Benefits of cross-verification

Before we describe the methodology we have developed to cross-verify the Compustat data, let us first discuss the benefits of such procedure and explain how it can be helpful for empirical accounting research that utilizes Compustat.

Cross-verification in our context means that for each accounting number of interest reported in Compustat, we try to find the corresponding number as reported by the company in its financial report. Then we compare each pair of such numbers to see if there is any difference between them. If there is a difference, we try to explain this discrepancy by analyzing it.

There are several immediate benefits of cross-verification. First of all, we can mark all Compustat data items for which there were no discrepancies found as accurate numbers. These numbers agree with companies' financial reports, and hence are to be trusted. Such numbers should not be excluded from the sample or transformed even if they are extreme, unless the scope of the study specifically excludes extreme observations. In fact, "black swan" observations are at least as important as other observations. Also, empirical models built using accurate cross-verified numbers would be more credible than those that use unverified numbers.

The second immediate benefit of data cross-verification is that researchers can estimate the amount and magnitude of Compustat intentional alterations of data made to fit Compustat's standard definitions of variables. Some of these alterations result in millions and billions of U.S. dollar differences. E.g., Compustat increased reported Revenues of AIG (American International Group) in 2011 by \$116,000,000, and decreased Amazon.com's reported Cost of Goods Sold by \$1,000,000,000. As noted previously, some Compustat's intentional data alterations are questionable, and accounting researchers that rely on data reported in Compustat need a better understanding of these alterations and their impact.

Finally, cross-verification allows to identify data items in Compustat that are either adjusted numbers, erroneous numbers, non-updated numbers, or missing values. Researchers

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should exercise extreme caution when utilizing such data items in their studies.

An important additional step in the cross-verification procedure in this study is the analysis of discrepancies. By analyzing each discrepancy, it is possible to get an accurate data population (or sample). Analysis of discrepancies is the only way to ensure that the data utilized is correct and represent the underlying business facts to the degree accounting numbers allow it.

It should be noted, however, that cross-verification using financial reports may not be enough to check all data items reported in Compustat. For example, many U.S. GAAP companies (approximately 30%) do not report directly such a fundamental accounting number as Total Liabilities in their financial reports – Total Liabilities are often reported together with Stockholder's equity as one number. Therefore, additional calculations are needed to get the amount of Total Revenues. An advantage of using the Compustat data set, is that Compustat calculates many accounting numbers that are not directly reported in financial statements. However, verification of those numbers may not be an easy task due to the complex nature of calculations of some accounting items.

4.2 XBRL as a means for corporate accounting data verification

In our study, we take advantage of the emerging XBRL financial reporting to cross-verify the annual data items reported Compustat. In essence, XBRL is a formal language for communicating business information. Data items are described using meta-information, and are linked together through various relationships. Information carried by XBRL is both human- and computer-readable. A special software can render an XBRL document and present it to an end-user as an electronic document humans are used to dealing with. However, the real advantage of XBRL is that it allows computer software to parse XBRL documents, and find, extract, and present information in an automated fashion. This drastically reduces the cost of manual labor needed to process such documents. It also allows to process a large amount of documents within a matter of seconds.

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The SEC has realized the benefits of XBRL reporting, and in April 2005 the SEC adopted *XBRL Voluntary Filing Program* (SEC 2005). This program enabled U.S. GAAP filers to voluntarily prepare financial statements using XBRL. The objective of the program was to determine the usefulness of XBRL as a format for reporting financial information by analyzing volunteers' feedback comments. In 2009, the SEC adopted *Interactive Data to Improve Financial Reporting* final rules (2009a) that mandated companies that prepare their financial statements in accordance with the U.S. GAAP to file supplemental XBRL documents for 10-Q, 10-K, and 8-K reports in addition to the plain-text ones. This mandate was implemented in several phases. In the first phase, domestic and foreign large accelerated filers with worldwide public common equity float above $5 billion and the fiscal periods ending on or after June 15, 2009, were required to file with XBRL. In the last phase, all the U.S. GAAP companies were required to file using XBRL for the fiscal years ending on, or after June 15, 2011.

One of the most fundamental uses of corporate accounting data is for financial data comparison. Users of financial statements need to compare financial positions of various companies for their decision-making. Since XBRL is just a language that can be used for business data reporting, it does not have built-in capabilities for multi-document data comparison. Therefore, the SEC has created the XBRL U.S. GAAP Financial Reporting Taxonomy. This taxonomy is a collection of common accounting data concepts, definitions, types, and relations that is meant to accommodate most of companies' financial reporting needs. Since most accounting items in financial reports is fairly standard, the XBRL U.S. GAAP Financial Reporting Taxonomy defines common rules how to present this standard information in XBRL filings. Therefore, in theory, standard accounting numbers can be easily compared across different filings using the XBRL U.S. GAAP Financial Reporting Taxonomy. Accounting information that is not standard, i.e., company- and filing-specific information, is represented in XBRL documents through so-called extensions. Extensions are an imported part of XBRL filings that provide additional reporting flexibility.

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As mentioned above, public U.S. GAAP companies are required to file their financial reports using the XBRL reporting technology starting from June 15, 2011, with standard accounting data items being formally reported in a similar manner due to the use of common XBRL U.S. GAAP Financial Reporting Taxonomy. Given that accounting variables found in Compustat are fairly standard, it becomes clear that XBRL filings can be utilized to automatically extract standard accounting numbers from corporate financial reports and compare them to numbers found in Compustat.

An issue that should be mentioned separately is the data quality in XBRL filings. Ideally, XBRL fillings should be as credible as plain-text filings themselves. Unfortunately, due to the complicated nature of XBRL and idiosyncrasies of the SEC's implementation of the XBRL U.S. GAAP Financial Reporting Taxonomy, XBRL filings are not error-free (Debreceny et al. 2010; Bartley et al. 2011; Du et al. 2013). XBRL filings are not audited in the U.S. at the moment, and do not replace the main plain-text filings. As of now, XBRL filings enjoy limited liability for the first 24 months after the filer was first required to submit XBRL filing, but no later than October 31, 2014 (SEC 2009b). This means that for the most filers, the limited liability clause has expired in June 2013. This may indicate that recent XBRL filings should be more accurate, and that XBRL filings will become more accurate in the future.

The reason for the XBRL reporting limited liability provision is that filing using XBRL is a complex undertaking, and filers may not be able to produce accurate XBRL reports immediately. There is a certain learning curve associated with XBRL reporting. As preparers of financial statements gain more experience with the XBRL technology, XBRL financial reports should become more accurate. The effect of this learning curve complements the effect of having a more substantial liability in the future. Hence, it is reasonable to expect XBRL reporting becoming more accurate in the future.

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4.3 Cross-verification methodology

In our study, we compare annual accounting data of 5014 company-unique XBRL filings to the appropriate Compustat data for the period from October 1, 2011, to September 30, 2012. There are several reasons why we have chosen this particular period. First of all, to better assess the data quality in Compustat we wanted to analyze only one annual filing per company. This ensures that we do not introduce additional bias and noise to the data and cover all possible companies and industries. Secondly, we analyze filings for the recent 2011-2012 period to capture the current data quality state of the Compustat data set. Moreover, a more recent period makes more sense from the XBRL perspective since annual XBRL filings are a recent addition to corporate 10-K filings, and, as noted previously, recent XBRL filings are likely to be more accurate than the older ones. Finally, we downloaded the data on February 22, 2013 which introduces a 145-day lag between the latest reporting period and the data download date. This ensures that most of the late filings will be captured in our data, and ensures that Compustat had sufficient amount of time to update missing and restated data items.

We identify six steps in the data cross-verification process:

1. Extract data from Compustat.
2. Extract data from XBRL 10-K filings.
3. Merge Compustat and XBRL data.
4. Create mappings between Compustat variables and XBRL reporting concepts.
5. Calculate differences between Compustat variables and the associated XBRL reporting concepts.
6. Analyze discrepancies between Compustat and XBRL 10-K filings.

Below, we explain each step in more detail.

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4.3.1 Extracting data from Compustat

One of the main advantages of using Compustat is that the data extraction process is fairly simple. This feature is particularly attractive to researchers that rely on accounting numbers in their studies. However, it may be the case that many users of Compustat do not realize that some accounting numbers reported by companies have been altered to fit Compustat's definitions of variables. It is important to understand the data alterations and the effects they may have on the output of a research that relies on the Compustat data.

We extracted all the available variables and their footnotes from Compustat Fundamentals Annual through Wharton Research Data Services (WRDS) interface for the U.S. companies for the period from October 1, 2011, to September 30, 2012. If there was more than one observation for a company, we kept only the latest one.

Dollar values in XBRL filings are measured in $1 denominations. However, in the Compustat data set, different variables use different units of measurement (numéraires). For example, the Net Income variable is measured in millions of U.S. dollars ($1,000,000), while Earnings per Share (EPS) is measured in U.S. dollars ($1). We used variable definitions in Compustat Manual to transform all Compustat values to be measured in U.S. dollars ($1).

4.3.2 Extracting data from XBRL 10-K filings

We downloaded all the domestic XBRL 10-K filings (including amendments) from the SEC's Electronic Data-Gathering, Analysis, and Retrieval (EDGAR) system with the reported fiscal year being in the one-year range from October 1, 2011, to September 30, 2012. The filings were downloaded on February 22, 2013. Hence, most of the late filings and filing amendments are likely to be included in the data set. For each company, we kept only the latest 10-K filing, and, where applicable, all of its amendments. We kept only the latest company filing to ensure that the reported amount and magnitude of discrepancies are not biased by certain types of companies and data alterations. And the reason to have both amendments and the original filings is to verify whether Compustat updates original

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numbers to restated ones in a timely fashion.

XBRL 10-K filings use the XBRL U.S. GAAP Financial Reporting Taxonomy, a collection of common financial and reporting concepts, definitions, types and relationships. Although, filers can define their own XBRL data concepts, SEC encourages them to use standard concepts found in the XBRL U.S. GAAP Financial Reporting Taxonomy whenever possible. By doing this, filers in their XBRL fillings would report similar accounting concepts using similar XBRL U.S. GAAP elements, which would greatly enhance data comparability across different companies and filings. Because of this, we extracted only those reporting concepts that are defined in the XBRL U.S. GAAP Financial Reporting Taxonomy.

Unfortunately, extracting data from XBRL documents is not a simple process. The main problem is that filers do not utilize XBRL reporting technology in a consistent manner. This is partially caused by the amount of flexibility that the SEC allows in XBRL reporting, partially by the limited guidance on how to properly use the XBRL U.S. GAAP Financial Reporting Taxonomy, and partially by the filers' negligence in preparing XBRL documents. The end result is that even though filers use the same XBRL U.S. GAAP Financial Reporting Taxonomy, different filers describe similar standard accounting items differently in their XBRL reports. As a consequence, it is quite a challenge to extract accurately the information of interest from many filings in an automated fashion.

A particularly problematic current XBRL reporting practice is the use of XBRL contexts. An XBRL *context* provides additional information about the concept reported, usually including such information as the concept's associated period of report, entity, dimension, etc. Although a useful instrument, companies tend to use it differently, thus reducing the benefit of XBRL reporting standardization. For example, the XBRL U.S. GAAP Financial Reporting Taxonomy defines "LegalEntityAxis" dimension that identifies the specific entity that a concept is related to. For an XBRL concept associated with a parent company, different fillings may provide: 1) no value for the XBRL "LegalEntityAxis" dimension, 2) "ParentCompanyMember" value for the "LegalEntityAxis", or 3) a custom member definition for the

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"LegalEntityAxis" dimension. This example shows two major issues. The first one is that it is not clear what context should a concept of interest have. I.e., there can be many instances of the same concept, each having its own context; the problem is to pick the correct one. The other issue is the concepts' entity attribution – in many cases, a filing reports accounting information for many entities, and it may be problematic to attribute each concept to its corresponding entity since each entity may have its own value for the "LegalEntityAxis" dimension that is usually an extension (i.e., not defined in the XBRL U.S. GAAP Financial Reporting Taxonomy).

Theoretically, if one filing is associated with several U.S. GAAP filers that have Central Index Keys (CIKs) assigned to them, the XBRL filing may include information about those filers through the means of special "Document and Entity Information" (DEI) XBRL concepts. However, in practice, this is rarely the case, and not much information is revealed about the entities other than the main one in the XBRL filing.

The issues described above make the processes of extraction of relevant information and its attribution to various entities very complicated and perhaps even ambiguous. It is true that those issues are irrelevant when a user of financial statements is concerned with only one filing – an XBRL software can render the filing in a comprehensible format. However, the same functionality is provided by a plain-text filing without the cost of utilizing the XBRL reporting technology. The main advantage of XBRL is that computer software can automatically process the embedded tags in the XBRL document without any need for user intervention. This is especially important when a software processes many XBRL documents and compares the data within those documents. Hence, the value of XBRL decreases when filers become too frivolous in the way they utilize XBRL.

To overcome the problem of attributing a filing's reporting concepts to entities associated with that filing, we need to identify the correspondence between the values of the XBRL "EntityLegalAxis" dimension embedded in the concept contexts and the actual entities. In this study, we utilize the following method:

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1. Use the EDGAR index file to extract information about entities, such as legal names, Central Index Keys (CIKs), addresses, etc.

2. Whenever possible extract entity context information from "Document and Entity Information" (DEI) XBRL concepts embedded in the XBRL instance document.

3. Look for standard entity member values such as "ParentCompanyMember", "SuccessorMember", etc. in the XBRL instance document.

4. Extract descriptions (labels) for all possible values of the "EntityLegalAxis" dimension found in the XBRL label linkbase document.

5. Match entities' information found in the EDGAR index files with the values found in the XBRL filing.

The last step involves comparing the values of CIKs (whenever possible), or the descriptions (labels) of the XBRL entity member values with the entities' legal names. Since the entity names are often not exact, e.g. "Microsoft Corporation" and "Microsoft Corp.", we use the Jaccard text similarity measure (Jaccard 1901) to compare labels and EDGAR legal names. The Jaccard similarity measure is defined as follows. Formally, given two pieces of text, let T_1 be the set of all words in the first text, and T_2 be the set of all words in the second one. The Jaccard similarity measure between these two pieces of text is defined as

$$J(T_1, T_2) = \frac{|T_1 \cap T_2|}{|T_1 \cup T_2|}. \tag{1}$$

That is, the Jaccard similarity measure is equal to the number of shared words divided by the number of all words.

Finally, by means of their contexts, XBRL concepts may be defined in other dimensions than the entity dimension, such as geographical area. Since we want to compare XBRL data items with Compustat data items that are usually general in nature, we do not extract the concepts that are defined in dimensions other than the entity dimension. The only exceptions

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are dimensions that indicate a restatement, type of financial report, or components of partner capital.

For the purpose of our study, we created our own applications that would process XBRL 10-K filings due to unavailability of such software in the public domain. However, recently a number of tools have appeared that are able to process a large number of XBRL filings. Many of these tools are still in the development stage, and may not be fully functional. However, potentially they can be used by academics and practitioners to cross-verify Compustat and other data aggregators' data sets to obtain data of the highest quality.

4.3.3 Merging Compustat and XBRL 10-K data

Data items in Compustat and XBRL 10-K observations should be merged based on the same entity and reporting period. Matching data by reporting periods does not present a particular problem. However, matching by entities does.

Each company in the EDGAR database is identified by the SEC's unique identifier – Central Index Key (CIK). In addition, EDGAR's XBRL filing document name convention requires to identify the main entity through its Ticker code, a unique identifier for stock market public traders. Compustat has its own unique identifier called Gvkey. However, Compustat does provide values for both CIK and Ticker.

The CIK is the best unique entity identifier to match data items from both data sets since EDGAR may not provide Ticker codes for all entities associated with a filing. However, for many observations, the value of the CIK variable in Compustat is missing. In those cases, we use either 1) the combination of Ticker code, company name and address, or 2) the company name and address to match the entities across data sets. As before, since the company name and address may not be represented uniquely (e.g. "One Green Avenue" versus "1 Green Ave."), we use the Jaccard text similarity measure (described in §4.3.2) to match the appropriate text fields.

Overall, we extracted 7,466 XBRL 10-K company-unique observations, and 7,375 Com-

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pustat company-unique observations for the period from October 1, 2011, to September 30, 2012. Out of those, we were able to merge 5,014 observations. Most of the companies that we were unable to match were either 1) too small to be present in Compustat 2) not present in the EDGAR database, or 3) did not file 10-K with XBRL.[15]

4.3.4 Creating mappings between Compustat variables and XBRL reporting concepts

To calculate differences between Compustat and XBRL 10-K data sets we need to match each Compustat item to the appropriate XBRL reporting concept. It is important to note that this is not a one-to-one match. For example, a Compustat variable "Depreciation and Amortization" can be represented in the XBRL U.S. GAAP Financial Reporting Taxonomy either as "Depreciation," "DepreciationAndAmortization," or "DepreciationDepletionAndAmortization" reporting concept.

Creating a mapping between Compustat variables and XBRL manually would be very costly and, most likely, inefficient. For this study, we developed an algorithm that creates mappings between variables in the data sets in an automated fashion. The algorithm is implemented under the assumption that Compustat variables and the corresponding 10-K XBRL concepts should have similar values in most cases when the values in Compustat are not altered. In other words, we assume the number of discrepancies between the two data sets to be rather small for the unaltered data. For each Compustat variable, the algorithm identifies a set of XBRL U.S. GAAP concepts that may match the variable using the information from the merged data set. The algorithm does not use XBRL extension reporting concepts because XBRL extensions are: 1) unique to each filing, 2) relatively rare (around 12% according to Du et al. (2013)), and 3) usually refer to non-standard concepts that are not present in Compustat.

15. The SEC's mandate to file using XBRL as of June 15, 2011, required to file all financial reports starting with the first 10-Q. That is, if for some company the first filing after June 15, 2011, was not 10-Q, but 10-K, the company was allowed not to file this 10-K using XBRL. For more details see item 601, Regulation S-K.

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The mapping algorithm works as follows. For each Compustat variable, it identifies all observations from the merged data set where that Compustat variable has a non-zero value.[16] Then, for each of those observations, the algorithm finds all the U.S. GAAP XBRL concepts whose values are equal to the one of the Compustat variable (up to a rounding error), if there are any. XBRL concepts that match the Compustat variable in at least 10% of all cases are considered to be *mapping candidates* for that Compustat variable. Ideally, a Compustat variable should match at least one of its mapping candidates in all the observations. Finally, the algorithm assesses how well the mapping candidates fit a Compustat variable, by calculating a measure that we call *mapping coverage*. Mapping coverage is defined as the number of observations where Compustat variable was matched by one of the XBRL mapping candidates divided by the number of all observations with that Compustat variable. For example a mapping coverage of 0.5 means that the obtained variable mapping explains at least 50% of Compustat values for that variable. Most variables that we used in our analysis have relatively high values of mapping coverage, but some do not due to large amounts of Compustat data alterations.

4.3.5 Calculating differences between Compustat variables and the associated XBRL reporting concepts

Once the data sets have been merged and the mappings between Compustat and XBRL 10-K items have been established, it is possible to calculate the data differences between the two data sets.

For each Compustat variable, we compare values of that variable to values of the appropriate XBRL concepts as defined by the mappings that we have previously established. If a value of a Compustat variable matches the value of at least one matching XBRL concept, then we assume that there is no difference between the Compustat value and the XBRL 10-K value. In general, when we compare values of Compustat and XBRL 10-K filings, three

16. Although, Compustat data allows variables to have no value, we have found that sometimes zero values also indicate no value.

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cases are possible:

1. There is no difference between values.

2. There is a difference between values.

3. Either Compustat or XBRL value is missing.

It is not clear why the Compustat data set would have missing values apart from the case where the values are not applicable or not reported by a filer. From the XBRL perspective, if Compustat reports a value and XBRL 10-K filing does not report matching concepts, then this is an artifact of either Compustat reporting calculated values, shortcomings of the variable mapping process, or errors in the XBRL filing. Unfortunately, we found that there are a number of issues with XBRL 10-K reports that result in missing values. Some typical problems are:

- Accounting number is not reported in an XBRL filing.
- Accounting number is reported using a wrong XBRL U.S. GAAP concept.
- A standard accounting number is reported through the use of extension.
- Accounting number refers to a wrong period.

In many cases, there is a difference between the values reported in Compustat and XBRL 10-K. We express this difference as a relative difference. We utilize the relative difference measure as opposed to the absolute one since it better reflects the magnitude of differences and allows to compare the extent of differences between different Compustat variables.

In cases where there is just one matching XBRL concept with a non-zero value, the calculation of the relative difference between Compustat and XBRL data items is straightforward. However, there may be cases where there are more than one matching XBRL concepts with different non-zero values. In such cases, we choose the XBRL concept whose value differs from the Compustat value the least, and calculate the relative difference between that XBRL

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concept and the Compustat value. We call such difference a *conservative relative difference*. Informally, it is equal to

$$\frac{\text{Minimum absolute difference between Compustat and matching XBRL values}}{\text{Compustat value}}. \quad (2)$$

Formally, if c is the value of the Compustat variable, and $x_1, x_2, \ldots, x_n$ are the values of the matching XBRL concepts, the conservative relative difference is calculated as:

$$\frac{\min\limits_{i=1\ldots n} |x_i - c|}{c}. \quad (3)$$

The conservative relative difference is a generalization of the regular relative difference – i.e., if there is only one matching XBRL value, than expression (3) is equivalent to the regular relative difference.

4.3.6 Analyzing discrepancies between Compustat and XBRL 10-K filings

Since we aim to get a better understanding of the data quality in Compustat, the mere calculation of differences between Compustat and XBRL 10-K data sets is not sufficient. We also have to investigate the reasons for these differences since sometimes they are products of the XBRL reporting. We utilize both automated and manual techniques described below to identify causes of discrepancies.

In §3, we discussed the possible reasons for a Compustat value to differ from the respective value in a financial report. However, XBRL 10-K reports may (and do) also contain errors.[17] For example, values may be reported without minus signs making them positive when they should be negative, or values may be reported in thousands of dollars instead of dollars. These will result in discrepancies. In addition, XBRL reports may use extensions or non-

17. For this study, the most important errors are errors that result in wrong values of wrong XBRL concepts. These may affect the results of our study, and are the ones that we try to eliminate. However, there may be other errors in XBRL data that do not affect the results of our study. For more information on 10-K XBRL quality, please see Hoffman (2013) and Boritz and No (2008).

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conventional dimensions to describe the data items. The algorithm we utilize to extract and match XBRL data may not be able to capture such XBRL items. If this happens, then in the best case it will result in a missing XBRL value in our matched data. In the worst case, it will result in a difference in the matched data set since the algorithm could have extracted other XBRL items that could sometimes match a Compustat variable (i.e., mapping candidates, see §4.3.4).

Given that discrepancies between data sets may exist due to XBRL-related errors, it is necessary to eliminate all such discrepancies from our analysis in order to assess the true amount and magnitude of data alterations in Compustat. Fortunately, it is possible to identify sources of most discrepancies in an automated fashion. For example, a simple check against a Compustat value may help identify if an XBRL value has a wrong sign. In addition to simple checks that compare data values against each other, we employ other automated procedures described below that target specific sources of discrepancies.

To find out whether the extraction algorithm was unable to extract the correct value from XBRL 10-K due to the value being reported in a non-standard manner or with an error (e.g. extension, wrong tag name, non-conventional dimensions, etc.), we employ another algorithm that automatically scans the XBRL 10-K filing to see if any other XBRL concept matches the Compustat value. If there is such a concept, the algorithm reports the reason for the XBRL concept being not extracted or matched. For example, the standard accounting item may be reported through the use of XBRL extensions, or the item may be attributed to a different entity.

As mentioned previously, Compustat sometimes fails to update numbers after a restatement has been filed. If the original 10-K filing was filed using the XBRL reporting technology, it is possible to check whether a discrepancy between Compustat and XBRL filing is due to the Compustat value being not up to date by comparing the Compustat value to the matching XBRL values in the original, pre-amendment 10-K filing.

Finally and most importantly, Compustat often alters original numbers reported in cor-

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porate filings to standardize values according to Compustat's definitions. It would be very costly to identify such alterations using manual procedures. Therefore, we developed an algorithm that tries to explain a difference between Compustat and XBRL 10-K filing by searching the XBRL filing for concepts whose values could explain the difference (up to a rounding error). Essentially, the algorithm tries to find the adjustments made to the original 10-K number by Compustat. The algorithm first tries to find an XBRL concept in the XBRL 10-K filing that if subtracted or added to the matching XBRL value results in no discrepancy with the Compustat value. If no such item is found, the algorithm tries to find a combination of two XBRL concepts and combination of subtractions/additions that would result in no discrepancy. Finally, if no such combination was found, the algorithm will search for a combination of three XBRL concepts and their respective combination of subtractions/additions that would result in no discrepancy. If no such ternary combination is found, the algorithm stops.

It should be noted that although the algorithm described above is very powerful at explaining Compustat data standardization, it may yield a spurious combination of complementing XBRL items - i.e., a combination that by chance results in adjustments that yield no discrepancies between Compustat and XBRL items. Therefore, it is imperative to identify such spurious combinations. For each variable, and each automatically found combination, we manually scan the adjustment and assess whether it appears suspicious or not. For example, it is very likely that Compustat may adjust the value of Cost of Goods Sold by adding the value of Labor and Related Expense item; however, it is highly unlikely that Compustat will adjust the value of Cost of Goods Sold by adding the value of Current Assets. The latter incident must be investigated manually by comparing the Compustat's value to the plain-text 10-K's value. Although we do apply our best judgment when deciding whether an automatically-found adjustment should be investigated manually or not, this is still an ambiguous process, and hence is a limitation of our study. On the other hand, we are interested only in those spurious combinations that help us eliminate XBRL errors, and

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we have found that the number of XBRL errors among these is very low (around 0% to 3%, depending on the variable).[18]

Many discrepancies may be explained in an automated fashion as described above, but some may not. In these cases, manual checks have to be performed. To do that, we compare values found in Compustat and XBRL 10-Ks to the values in the corresponding plain-text 10-K reports. It should be noted that the nature of discrepancies that require manual checks is such that it may be impossible to identify the exact reasons for having these discrepancies. However, we can usually tell whether the discrepancy is due to adjustment/error in Compustat, XBRL 10-K report, or both. Overall, we have manually cross-verified 1,800 discrepancy items (around 1.5% of all items) using original 10-K filings.

5 Analysis and results

For our analysis of differences between Compustat and 10-K numbers, we have chosen 30 top-level Compustat variables that are very common in financial reports and that are frequently used in accounting academic literature. For the full list of variables, their description and descriptive statistics, please see Table 2. Half of the analyzed variables are Balance Sheet variables, 11 variables are Income Statement variables, and 4 variables are Cash Flow Statement variables.

For all 30 Compustat variables we have performed the cross-verification procedure described in §4, i.e., we have merged the Compustat and XBRL 10-K data, created mappings between data sets, calculated differences and analyzed discrepancies. Most values from Compustat have been matched to the appropriate XBRL 10-K values. However, some were not; this is mostly due to a 10-K not reporting a number that can be calculated from other 10-K numbers, or due to the use of XBRL extension concepts. It is interesting to note from Table 2 that in some cases Compustat did not report values that were present in the 10-Ks (e.g.,

18. Most XBRL errors are usually found by applying other checks described in the section that targets XBRL errors specifically.

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Compustat did not report Total Assets values for 42 companies in our sample). Although, the number of missing observations is not very large, there should not be cases like these in a high-quality data set.

5.1 Discrepancy analysis by variables

Table 3 reports the amount and magnitude of discrepancies for each Compustat variable. As mentioned before, we define a discrepancy as a difference between Compustat and XBRL 10-K numbers that exceeds a rounding error. In addition, we adopt a 0.5% rule-of-thumb definition of material discrepancies – a Balance Sheet discrepancy is "material" if it exceeds 0.5% of Total Assets, an Income Statement discrepancy is "material" if it exceeds 0.5% of Net Income, and a Cash Flow Statement discrepancy is "material" if it exceeds 0.5% of reporting periods' change in Cash amount. These definitions of materiality are commonly used in practice and were utilized in Boritz and No (2013). In addition, we consider an Earnings per Share discrepancies to be material if the absolute difference between data set values exceeds 5 cents.[19]

In Table 3, we report two types of discrepancy statistics with mean, median, and standard deviation information. The first one summarizes difference information about observations that resulted in discrepancies, i.e., this is descriptive statistics of discrepancy observations. For each variable, we perform a one-tailed t-test to assess whether the average relative discrepancy is significantly larger than 0. This test allows us to assess whether values of discrepancies are significant. For 22 (out of 30) variables, we find that discrepancies are significant with the significance level of at least 90%. Since the mean value can be driven by large discrepancies, we also report the median value of discrepancies to have a better understanding of the "middle" discrepancy.

The second type of descriptive statistics summarizes difference information about all

19. Some audit practices set the amount of materiality to the level that would change the value of EPS by 1 cent. We decided to take a more conservative (with respect to deciding what a discrepancy is) approach and define a discrepancy in EPS to be material if it exceeds 5 cents.

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matched Compustat and XBRL 10-K observations. This statistics is meant to show the effect of discrepancy observations on the whole population of observations. Here, the mean value signifies the average relative difference between all Compustat and XBRL 10-K matched observations. For each variable, we test whether there is a significant difference between Compustat and XBRL 10-K numbers by running a one-tailed t-test that determines whether the mean relative difference is significantly larger than 0. The results of these tests show that values of 22 (out of 30) variables significantly differ across Compustat and XBRL 10-K data sets (with the significance level of at least 90%). In addition to the mean values, we also report the median values of relative differences between Compustat and XBRL matched population, but since median is the "middle value", these values are equal to 0 unless the number of discrepancy observations for a variable exceeds the number of non-discrepancy observations.

The results in Table 3 clearly show that there are significant differences between Compustat and 10-K numbers. However, there are a couple of interesting observations that we would like to point out. Firstly, the amount of discrepancies, their magnitude, and their overall effect on the non-discrepancy observations differs by variable. High-level variables that have simple definitions (e.g. Total Assets = all assets, Total Liabilities = all liabilities, Net Income = all revenues - all expenses) tend to have less discrepancies than variables that have more complex definitions (e.g. Cost of Goods Sold, or Gross Profit). Since components of more complicated variables are harder to define precisely, they tend to be more adjusted by Compustat.

Secondly, Compustat provides two variables of Retained Earnings item that we include in our analysis – Retained Earnings (RE) and Retained Earnings Unadjusted (REUNA). The difference is that the former is the Compustat-adjusted (standardized) version of Retained Earnings, while the latter should represent numbers as reported on 10-K without any adjustments.From Table 3 it follows that the values of REUNA do differ from XBRL 10-K values in 1.36% of all cases (with 0.97% being material). However, what is even more striking

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is that RE values differ from XBRL 10-K values in 72.2% of all cases (with 62.54% being material) and that these discrepancies result in the average relative difference of 17.65% between all matched Compustat and XBRL observations. Such a large difference between REUNA and RE variables cannot be an artifact of the XBRL reporting or due to limitations of our study (as discussed in §4.3.6) since REUNA values have matched XBRL 10-K values in 98.64% of all cases. This important example indicates that Compustat heavily adjusts original numbers reported on 10-Ks.

Finally, as already mentioned, for most variables the mean relative differences between Compustat and XBRL 10-K numbers are statistically significant. This means that the distributions of accounting variables in Compustat and 10-Ks are significantly different which may have a profound effect on the results of accounting studies that utilize the Compustat data set. This, in particular, is true for studies that utilize linear regression models since these models are very sensitive to changes in data distribution (Klein and Rossin 1999).

Our results agree with the results of Tallapally et al. (2011); Tallapally et al. (2012), that compared Cost of Goods Sold and Sales values between Compustat and XBRL 10-K filings for a sample of 27 companies, in a sense that the Cost of Goods Sold item tends to have more discrepancies and with larger magnitudes than the Sales item. Some of our results differ slightly from the results of Boritz and No (2013). For example, we do not find Total Liabilities to be significantly different across the data sets (both in number and magnitude), but we do find higher rate of discrepancies for the Receivables item (17% versus 7.3%). The differences are most likely caused by the fact that we study a much larger sample and have only one observation per company in our sample.

5.2 Discrepancy analysis by industry, size, and XBRL adoption phase

Company characteristics such as industry and size affect what is reported in 10-K statements and how it is reported. The same items will often have different contexts for different

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companies – e.g., things that comprise Cost of Goods Sold item of a retailing company are very different from components of Cost of Goods Sold item of a manufacturing company. Therefore, it is important to study the amount and magnitude of discrepancies with regard to different characteristics of companies. In our study, we look at three company characteristics: industry, size and XBRL adoption phase.

There are several ways to define company industry (e.g. using SIC or NAICS codes). In this study, we utilize Fama/French 12-industry classification system (Fama and French 1997) since it has been used extensively in accounting literature. In addition, there are only 12 industry groups (as opposed to 75 groups defined by 2-digit SIC codes) that facilitates better delivery and comprehension of results. Please refer to Table 1 for the list and description of Fama/French 12 industries.

As has been mentioned previously, in 2009, the SEC adopted *Interactive Data to Improve Financial Reporting* final rules that mandated U.S. GAAP companies to report their financial statements with XBRL. The mandate was implemented in three phases. Phase one with the deadline in June 2009 required all large accelerated filers with worldwide public common equity float above $5 billion to adopt XBRL. Phase 2 with the deadline in June 2010 required all other large accelerated filers to adopt XBRL. Finally, Phase 3 with the deadline in June 2011 required all U.S. GAAP companies to file with XBRL. We use the XBRL adoption phase as a characteristic of a company – XBRL adoption phase can be thought as one measure of company size. In addition, we use company revenues as yet another, more refined, measure of size.

We provide discrepancy information by industry and XBRL adoption phase in Tables 4 and 5. Table 4 reports the number of discrepancies, and the median relative value of discrepancies for each Compustat variable and Fama/French industry.[20] In Table 5, we report discrepancy statistics by industry and by XBRL adoption phase. These statistics include information about discrepancy counts as well as mean and median relative discrepancies.

20. We report median values instead of means because the former are less biased statistics, i.e. they are not influenced by extreme values of discrepancies as much as mean values.

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Results in Tables 4 and 5 seem to suggest that there is a difference in amounts and magnitude of discrepancies for different variables, industries, and sizes. To test it formally, we utilize analysis of deviance of discrepancy observations. Analysis of deviance is somewhat similar to analysis of variance (ANOVA) and analysis of covariance (ANCOVA), but can be used for generalized linear models (instead of linear regressions). It allows to test whether there are significant differences between values of some response variable for different groups (categorical variables) while controlling for other continuous variables.

We utilize analysis of deviance in two ways. First, we test whether the fact of an observation being a discrepancy or not is related to company characteristics. In other words, we assume that an observation can have two states – either being a discrepancy or not – and test whether the value of that state depends on company characteristics. Since the response variable in this case is a state of an observation, this type of an analysis is an observation-level analysis. Second, we test whether company characteristics are related to the number of discrepancies for that company. For each company in the matched data set, we calculate the ratio of the number of discrepancy observations to the number of all matched observations for that company. The resulting ratio is a number between 0 and 1, with 0 indicating that there are no discrepancy observations for that company, and 1 indicating that all matched observations are discrepancies. Then, we test whether such company ratios depend on company characteristics. This kind of analysis is a company-level analysis since each company discrepancy ratio is defined for a company. Because response variables values are not unbounded real numbers – in the first case it is a binary variable, and in the second case it is a number between 0 and 1 – we cannot assume that they are normally distributed and utilize ANCOVA. However, we can use generalized linear models (logistic regressions) and analysis of deviance instead.

Analysis of deviance results are presented in Table 6. In addition to studying how company characteristics are related to discrepancies, we also study how these characteristics are related to material discrepancies. Hence, Table 6 shows results both when we use discrep-

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ancy notion for our response variables and when we use material discrepancy notion. In addition to company characteristics, in our analysis, we include the type of statement where the accounting variables are reported since Tables 3 and 4 suggest that the amount and magnitude of discrepancies also depend on the type of a variable.

Panel A of the table contains the results of deviance analyses that study the effects of revenue (size), industry, phase, variable statement, and various interactions between industry, phase, and variable statement. These results are fairly consistent and show that effects of all characteristics and their interactions are statistically significant except for the XBRL adoption phase in all cases, and revenue in one case. Although the main effect of XBRL adoption phase is not statistically significant, its interactions with other characteristics are. The insignificance of the main effect can be explained by XBRL adoption phase being related to revenue since they both represent the size of a company.[21] To avoid this effect, we conduct another set of analyses with the XBRL adoption phase variable removed. The results are reported in Panel B of Table 6, and show that all characteristics and their interactions are statistically significant, including revenue. Moreover, the significance of revenue has increased drastically.

Overall, the results show that the type of variable, industry, size, and XBRL adoption phase are related to the amount and magnitude of discrepancies. The results are slightly different from Boritz and No (2013) in a sense that we do find industry to be a significant factor (with significance levels of 99.9% in all cases) as related to the number of discrepancies.

6 Concluding remarks

In this study, we conduct the first large-scale comparison of Compustat and 10-K accounting numbers to study the amount and magnitude of data alterations in the Compustat data set. Specifically, we compare 30 accounting items commonly used in the accounting literature for

21. This was confirmed by the results of an additional ANOVA test (not reported in the paper) that showed the differences between mean values of revenues across different XBRL adoption phases being 99.9% significant. Phase 1 companies had on average the largest revenues, and phase 3 companies the smallest.

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more than 5,000 of U.S. GAAP companies for the period from October 1, 2011, to September 30, 2012. This large-scale comparison has become possible due to SEC-mandated XBRL financial reporting for all U.S. GAAP filers as of June 2011, since XBRL allows to extract data from XBRL 10-K filings in an automated fashion

We have developed and presented a methodology to cross-verify Compustat numbers with XBRL numbers. This methodology addresses issues of data extraction, merging, variable mapping, difference calculation, automated error detection, and discrepancy analysis. The cross-verification not only provides insights about data alterations in the Compustat data set, but also is a useful tool for accounting researchers to validate their data.

We find that Compustat significantly alters numbers reported in the 10-K filings. Specifically, we find that Compustat values of 22 (out of 30) variables significantly differ from values reported in 10-K filings. Variables that usually do not have significant amount and magnitude of discrepancies are variables that have fairly simple definitions (e.g. Total Assets, Total Liabilities or Net Income). Variables that have more complex definitions (e.g. Cost of Goods Sold or Gross Profit) are more likely to differ across the data sets.

We also find that the type of statement where variable is reported and company characteristics such as industry, size, and XBRL adoption phase are related to the amount and magnitude of discrepancies.

Our findings suggest that data alterations in Compustat may influence results of accounting studies that utilize the Compustat data set. Compustat itself argues that its data standardization improves data comparability across companies. However, empirical evidence presented in this paper suggests that this is not always the case. Therefore, more research and discussion are needed on this matter.

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Appendix

Table 1: Fama/French 12 industry classification

Industry Variable	*Variable Description*	*Examples of Industries Included*
NoDur	Consumer non-durables	Food, Tobacco, Textiles, Apparel, Leather, Toys
Durbl	Consumer durables	Cars, Television Sets, Furniture, Household Appliances
Manuf	Manufacturing	Machinery, Trucks, Planes, Office Furniture, Paper, Printing
Enrgy	Energy	Oil, Gas, Coal Extraction and Products
Chems	Chemicals	Chemicals and Allied Products
BusEq	Business equipment	Computers, Software, and Electronic Equipment
Telcm	Telecommunications	Telephone and Television Transmission
Utils	Utilities	Water, Gas, Electricity Utilities
Shops	Shops	Wholesale, Retail, and Some Services (Laundries, Repair Shops)
Hlth	Health	Healthcare, Medical Equipment, and Drugs
Money	Money	Finance

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Table 2: Descriptive statistics of Compustat variables

Statement	*Variable*	*Variable Description*	Observation counts			Descriptive Statistics of Matched Data		
			Available	*Missing*	*Matched (Perc.)*	*Mean*	*St. Dev.*	*Median*
Balance Sheet	ACT	Current Assets	3,946	0	3,902 (99.11%)	1,073.53	4,237.43	132.31
	CH	Cash	4,848	0	4,791 (99.07%)	374.64	2,899.25	26.70
	RECTR	Receivables (Trade)	3,849	0	3,414 (88.93%)	536.88	6,365.74	42.93
	INVT	Inventories	3,374	0	2,435 (72.26%)	330.71	1,317.23	36.44
	PPENT	Property, Plant and Equipment	4,582	0	4,381 (95.84%)	1,213.73	6,164.46	40.05
	DPACT	Depreciation, Depletion and Amortization	3,978	0	2,122 (54.61%)	1,621.78	7,357.35	231.16
	GDWL	Goodwill	2,659	0	2,541 (95.71%)	991.02	4,162.30	80.75
	AT	Total Assets	4,958	42	4,932 (99.74%)	8,581.13	84,465.52	509.03
	LCT	Current Liabilities	3,978	0	3,880 (97.76%)	737.45	3,153.04	59.06
	AP	Accounts Payable (Trade)	4,862	0	4,591 (94.66%)	1,883.37	30,076.04	23.54
	DLTT	Long-Term Debt	3,586	0	2,724 (76.17%)	2,174.16	12,320.60	237.90
	LT	Total Liabilities	4,965	0	3,540 (71.60%)	8,411.89	93,457.13	255.79
	RE	Retained Earnings	4,860	0	4,461 (92.28%)	802.44	8,134.33	9.43
	REUNA	Retained Earnings (Unadjusted)	4,714	0	4,329 (92.46%)	876.40	8,526.66	8.51
	TEQ	Stockholders' Equity	4,980	0	4,856 (97.80%)	1,652.20	8,795.99	145.95

Available – observations present in Compustat; *Missing* – observations present in XBRL 10-K filings, but not in Compustat.

Matched – observations present both in Compustat and XBRL 10-K filings excluding erroneous XBRL observations.

Mean, St. Dev., Median – values of descriptive statistics based on Compustat values of the matched observations.

Mean and median values are measured in millions of U.S. dollars, except for per share numbers that are measured in U.S. dollars.

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Table 2 : (continued from the previous page)

Statement	*Variable*	*Variable Description*	Observation counts			Descriptive Statistics of Matched Data		
			Available	*Missing*	*Matched (Perc.)*	*Mean*	*St. Dev.*	*Median*
Income Statement	REVT	Revenue	4,679	0	3,619 (77.73%)	3,489.00	15,996.30	359.45
	SALE	Sales/Turnover (Net)	4,679	0	2,410 (51.58%)	3,499.89	14,993.02	402.63
	IDIT	Interest and Related Income	1,973	0	1,600 (82.09%)	6.51	39.61	0.26
	COGS	Cost of Goods Sold	4,684	4	2,621 (56.11%)	2,129.08	10,312.62	164.34
	XAD	Advertising Expense	1,948	0	1,012 (52.16%)	124.25	497.74	5.15
	XINT	Interest and Related Expense	1,862	0	1,605 (86.85%)	147.38	688.97	16.10
	XRD	Research and Development Expense	1,861	0	1,601 (86.78%)	147.58	689.81	16.10
	GP	Gross Profit (Loss)	4,708	0	2,028 (43.08%)	783.94	3,418.41	97.46
	NI	Net Income (Loss)	4,972	34	4,920 (99.29%)	198.94	1,387.93	5.83
	EPSPI	Earnings Per Share (Basic)	4,657	0	4,435 (95.66%)	0.33	45.01	0.39
	EPSFI	Earnings Per Share (Diluted)	4,658	0	4,437 (95.67%)	0.33	45.00	0.39
CF Statement	OANCF	Operating Activities	4,967	0	4,910 (99.25%)	424.75	2,703.35	21.57
	FINCF	Financing Activities	4,882	0	4,826 (99.12%)	-225.77	9,147.81	0.01
	IVNCF	Investing Activities	4,793	0	4,727 (99.16%)	-186.48	9,372.46	-18.44
	CHECH	Cash and Cash Equivalents - Increase (Decrease)	4,909	58	4,841 (98.90%)	22.05	685.63	0.03

Available – observations present in Compustat; *Missing* – observations present in XBRL 10-K filings, but not in Compustat.

Matched – observations present both in Compustat and XBRL 10-K filings excluding erroneous XBRL observations.

Mean, St. Dev., Median – values of descriptive statistics based on Compustat values of the matched observations.

Mean and median values are measured in millions of U.S. dollars, except for per share numbers that are measured in U.S. dollars.

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Table 3: Difference statistics of Compustat and XBRL 10-K numbers

Statement	*Variable*	*Variable Description*	Observation counts			Discrepancy statistics			Difference between Compustat and 10-K		
				Discrepancies							
			Matched	*All (Perc.)*	*Material (Perc.)*	*Mean*	*Median*	*St. Dev.*	*Mean*	*Median*	*St. Dev.*
Balance Sheet	ACT	Current Assets	3,902	13 (0.33%)	9 (0.23%)	8.27%**	1.41%	16.13	0.03%**	0.00%	1.01
	CH	Cash	4,791	40 (0.83%)	23 (0.48%)	86.65%**	9.75%	239.18	0.72%**	0.00%	22.98
	RECTR	Receivables (Trade)	3,414	580 (16.99%)	549 (16.08%)	32.98%***	18.75%	42.27	5.60%***	0.00%	21.37
	INVT	Inventories	2,435	115 (4.72%)	111 (4.56%)	35.04%***	26.19%	28.35	1.66%***	0.00%	9.64
	PPENT	Property, Plant and Equipment	4,381	213 (4.86%)	203 (4.63%)	44.21%***	30.19%	38.95	2.15%***	0.00%	12.80
	DPACT	Depreciation, Depletion and Amortization	2,122	98 (4.62%)	92 (4.34%)	44.70%***	30.78%	39.77	2.06%***	0.00%	12.66
	GDWL	Goodwill	2,541	18 (0.71%)	6 (0.24%)	7.27%**	0.68%	15.23	0.05%**	0.00%	1.39
	AT	Total Assets	4,932	17 (0.34%)	15 (0.30%)	164.69%	1.70%	647.02	0.57%	0.00%	38.10
	LCT	Current Liabilities	3,880	20 (0.52%)	13 (0.34%)	3,273.67%	0.90%	14,605.82	16.87%	0.00%	1,048.76
	AP	Accounts Payable (Trade)	4,591	383 (8.34%)	356 (7.75%)	622.75%**	46.93%	6,199.09	51.95%**	0.00%	1,796.63
	DLTT	Long-Term Debt	2,726	628 (23.04%)	573 (21.02%)	617.82%	11.53%	13,815.95	142.33%	0.00%	6,632.32
	LT	Total Liabilities	3,540	50 (1.41%)	39 (1.10%)	1,374.34%	0.88%	9,669.96	19.41%	0.00%	1,149.35
	RE	Retained Earnings	4,462	3,222 (72.21%)	2,790 (62.53%)	24.13%***	2.94%	165.02	17.43%***	0.87%	140.64
	REUNA	Retained Earnings (Unadjusted)	4,329	59 (1.36%)	41 (0.95%)	106.82%**	0.84%	357.43	1.46%**	0.00%	43.19
	TEQ	Stockholders' Equity	4,856	213 (4.39%)	198 (4.08%)	243.70%	6.01%	3,123.11	10.69%	0.00%	654.53

Matched – observations present both in Compustat and XBRL 10-K filings excluding erroneous XBRL observations.

All (Perc.) – number (and percentage of matched observations) of discrepancy observations.

Material (Perc.) – number (and percentage of matched observations) of material discrepancy observations.

Discrepancy statistics – descriptive statistics of relative value differences between Compustat and XBRL 10-K observations with discrepancies.

Difference between Compustat and 10-K – descriptive statistics of relative value differences between all matched Compustat and XBRL 10-K observations.

***, **, * indicate significance of mean difference t-tests at 99%, 95%, and 90% levels respectively.

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Table 3 : (continued from the previous page)

Statement	*Variable*	*Variable Description*	Observation counts			Discrepancy statistics			Difference between Compustat and 10-K		
				Discrepancies							
			Matched	*All (Perc.)*	*Material (Perc.)*	*Mean*	*Median*	*St. Dev.*	*Mean*	*Median*	*St. Dev.*
Income Statement	REVT	Revenue	3,619	345 (9.53%)	330 (9.12%)	7.44%***	0.80%	22.91	0.71%***	0.00%	7.39
	SALE	Sales/Turnover (Net)	2,410	122 (5.06%)	120 (4.98%)	14.42%***	1.57%	27.42	0.73%***	0.00%	6.91
	IDIT	Interest and Related Income	1,600	36 (2.25%)	31 (1.94%)	1,824.73%	95.90%	9,371.78	41.06%	0.00%	1,412.72
	COGS	Cost of Goods Sold	2,621	2,229 (85.04%)	2,220 (84.70%)	24.55%***	5.78%	350.61	20.88%***	4.66%	323.43
	XAD	Advertising Expense	1,012	24 (2.37%)	21 (2.08%)	2,475.41%	47.74%	9,888.73	58.71%	0.00%	1,538.39
	XINT	Interest and Related Expense	1,605	55 (3.43%)	54 (3.36%)	42.92%***	8.04%	81.48	1.47%***	0.00%	16.87
	XRD	Research and Development Expense	1,601	51 (3.19%)	50 (3.12%)	42.07%***	6.78%	84.12	1.34%***	0.00%	16.61
	GP	Gross Profit (Loss)	2,028	1,756 (86.59%)	1,751 (86.34%)	23.14%***	8.28%	106.29	20.04%***	7.09%	99.22
	NI	Net Income (Loss)	4,920	45 (0.91%)	39 (0.79%)	252.44%	0.90%	1,630.73	2.31%	0.00%	156.09
	EPSPI	Earnings Per Share (Basic)	4,435	132 (2.98%)	103 (2.32%)	1,321.48%**	36.11%	6,758.45	39.33%**	0.00%	1,183.19
	EPSFI	Earnings Per Share (Diluted)	4,436	133 (3.00%)	104 (2.34%)	1,292.78%**	33.53%	6,735.06	38.76%**	0.00%	1,182.67
CF Statement	OANCF	Operating Activities	4,910	134 (2.73%)	127 (2.59%)	43.63%**	2.27%	222.85	1.19%**	0.00%	37.36
	FINCF	Financing Activities	4,826	40 (0.83%)	38 (0.79%)	81.94%*	4.59%	329.64	0.68%*	0.00%	30.55
	IVNCF	Investing Activities	4,727	82 (1.73%)	79 (1.67%)	119.20%**	4.92%	642.18	2.07%**	0.00%	85.50
	CHECH	Cash and Cash Equivalents - Increase (Decrease)	4,841	119 (2.46%)	114 (2.35%)	24.96%***	1.93%	93.45	0.61%***	0.00%	15.09

Matched – observations present both in Compustat and XBRL 10-K filings excluding erroneous XBRL observations.

All (Perc.) – number (and percentage of matched observations) of discrepancy observations.

Material (Perc.) – number (and percentage of matched observations) of material discrepancy observations.

Discrepancy statistics – descriptive statistics of relative value differences between Compustat and XBRL 10-K observations with discrepancies.

Difference between Compustat and 10-K – descriptive statistics of relative value differences between all matched Compustat and XBRL 10-K observations.

***, **, * indicate significance of mean difference t-tests at 99%, 95%, and 90% levels respectively.

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Table 4: Discrepancy amounts by industries

Variable	*Variable Description*	Perecentage of Discrepancies (Median Relative Discrepancy) by Industry											
		NoDur	*Durbl*	*Manuf*	*Enrgy*	*Chems*	*BusEq*	*Telcm*	*Utils*	*Shops*	*Hlth*	*Money*	*Other*
ACT	Current Assets	0.00%	0.86%	0.77%	0.43%	0.79%	0.66%	0.00%	0.00%	0.47%	0.18%	0.18%	0.85%
		–	(0.04%)	(1.50%)	(99.90%)	(4.67%)	(94.59%)	–	–	(49.95%)	(0.64%)	(50.17%)	(15.88%)
CH	Cash	0.00%	0.00%	0.51%	0.43%	1.57%	0.53%	0.00%	0.00%	0.47%	1.06%	2.31%	1.28%
		–	–	(58.92%)	(99.90%)	(0.07%)	(50.95%)	–	–	(54.92%)	(0.03%)	(45.88%)	(1.76%)
RECTR	Receivables (Trade)	11.40%	16.38%	11.22%	22.98%	11.81%	10.20%	14.81%	38.81%	12.38%	6.38%	7.99%	11.08%
		(5.28%)	(11.41%)	(7.36%)	(12.32%)	(4.49%)	(13.26%)	(7.80%)	(35.51%)	(15.51%)	(10.16%)	(77.23%)	(18.68%)
INVT	Inventories	1.32%	4.31%	7.91%	1.28%	3.94%	3.58%	2.22%	3.96%	0.70%	1.06%	0.18%	3.26%
		(34.84%)	(17.27%)	(24.70%)	(61.62%)	(7.73%)	(28.72%)	(30.55%)	(24.03%)	(64.71%)	(6.39%)	(28.32%)	(56.74%)
PPENT	Property, Plant and Equipment	4.82%	10.34%	5.10%	11.06%	3.15%	1.99%	2.22%	7.43%	3.50%	1.42%	2.93%	8.79%
		(17.16%)	(36.69%)	(6.52%)	(94.90%)	(15.05%)	(33.96%)	(3.37%)	(7.70%)	(24.26%)	(5.65%)	(63.57%)	(61.53%)
DPACT	Depreciation, Depletion and Amortization	2.63%	6.90%	2.81%	11.06%	1.57%	4.11%	2.96%	9.41%	4.91%	3.20%	0.89%	4.82%
		(200.00%)	(84.71%)	(68.36%)	(96.67%)	(199.97%)	(200.00%)	(200.00%)	(3.04%)	(13.06%)	(200.00%)	(149.89%)	(85.79%)
GDWL	Goodwill	0.44%	0.00%	0.77%	0.00%	0.00%	0.40%	0.00%	0.50%	0.00%	0.53%	0.80%	0.28%
		(0.10%)	–	(7.35%)	–	–	(4.90%)	–	(0.01%)	–	(17.83%)	(0.65%)	(50.31%)
AT	Total Assets	0.00%	0.00%	0.51%	0.43%	0.79%	0.53%	0.75%	1.49%	0.47%	0.35%	0.80%	0.71%
		–	–	(50.17%)	(99.90%)	(17.55%)	(50.80%)	(0.00%)	(7.00%)	(49.95%)	(2.19%)	(99.90%)	(19.94%)
LCT	Current Liabilities	0.00%	0.86%	0.77%	0.43%	0.00%	0.93%	0.75%	0.00%	0.47%	0.71%	0.09%	1.28%
		–	(66.23%)	(15.80%)	(99.90%)	–	(5.04%)	(0.08%)	–	(50.07%)	(0.01%)	(99.90%)	(3.93%)
AP	Accounts Payable (Trade)	7.89%	4.31%	4.34%	21.28%	11.81%	4.11%	11.11%	4.95%	7.94%	6.38%	9.14%	8.65%
		(24.61%)	(37.58%)	(65.17%)	(36.21%)	(53.16%)	(81.73%)	(76.10%)	(26.64%)	(19.89%)	(121.69%)	(54.14%)	(69.61%)
DLTT	Long-Term Debt	8.77%	4.31%	10.46%	9.36%	10.24%	8.48%	18.52%	15.35%	14.02%	8.69%	20.23%	11.06%
		(11.03%)	(1.04%)	(10.45%)	(1.79%)	(9.01%)	(20.61%)	(3.59%)	(2.42%)	(7.66%)	(11.25%)	(17.30%)	(7.52%)
LT	Total Liabilities	0.00%	0.86%	1.53%	1.70%	0.00%	0.40%	0.75%	1.98%	0.70%	1.24%	2.40%	2.13%
		–	(3.60%)	(37.09%)	(13.79%)	–	(1.62%)	(0.00%)	(3.08%)	(0.24%)	(15.22%)	(1.00%)	(2.66%)
RE	Retained Earnings	73.68%	74.14%	74.74%	43.40%	70.08%	70.73%	65.19%	58.91%	60.75%	55.14%	70.81%	56.60%
		(5.44%)	(7.60%)	(8.09%)	(2.65%)	(10.50%)	(1.48%)	(1.86%)	(2.41%)	(1.65%)	(0.40%)	(4.78%)	(2.16%)
REUNA	Retained Earnings (Unadjusted)	0.88%	0.00%	1.28%	1.28%	3.15%	1.46%	1.48%	0.00%	0.70%	1.95%	2.84%	2.55%
		(1.67%)	–	(0.00%)	(20.90%)	(68.79%)	(12.92%)	(61.09%)	–	(12.33%)	(0.57%)	(95.62%)	(78.20%)
TEQ	Stockholders' Equity	1.32%	3.45%	3.32%	5.11%	5.51%	3.18%	8.89%	20.79%	4.22%	4.79%	3.46%	3.83%
		(4.67%)	(24.40%)	(2.12%)	(27.96%)	(19.20%)	(11.20%)	(27.78%)	(2.19%)	(9.90%)	(19.15%)	(16.13%)	(12.07%)

This table reports two number for each industry and variable. The first (upper) is the percentage of all discrepancy observations relative to all matched observations in the industry. The second (lower) is the median relative discrepancy value for the industry. For information about industry variables, please see Table 1.

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Table 4 : (continued from the previous page)

Variable	*Variable Description*	Perecentage of Discrepancies (Median Relative Discrepancy) by Industry										
		NoDur	*Durbl*	*Manuf*	*Enrgy*	*Chems*	*BusEq*	*Telcm*	*Utils*	*Shops*	*Hlth*	*Money*	*Other*
REVT	Revenue	7.02%	3.45%	4.08%	11.06%	5.51%	3.44%	3.70%	2.48%	4.92%	3.55%	17.83%	2.98%
		(1.82%)	(0.19%)	(0.27%)	(3.06%)	(0.43%)	(1.27%)	(3.03%)	(0.73%)	(0.64%)	(1.91%)	(0.81%)	(0.68%)
SALE	Sales/Turnover (Net)	6.58%	3.45%	4.08%	2.98%	4.72%	2.38%	2.96%	0.00%	4.21%	3.01%	0.71%	2.27%
		(2.17%)	(11.73%)	(0.31%)	(6.04%)	(0.43%)	(2.47%)	(1.89%)	–	(1.43%)	(2.34%)	(48.41%)	(4.32%)
IDIT	Interest and Related Income	0.44%	2.59%	1.28%	1.28%	0.00%	1.59%	0.74%	0.99%	1.64%	1.24%	0.71%	1.56%
		(200.00%)	(200.00%)	(100.00%)	(99.90%)	–	(149.24%)	(125.98%)	(149.95%)	(200.00%)	(199.27%)	(77.96%)	(200.00%)
COGS	Cost of Goods Sold	74.12%	87.93%	87.50%	10.64%	77.95%	80.66%	37.04%	10.40%	53.97%	53.64%	5.06%	33.19%
		(4.63%)	(4.18%)	(4.33%)	(11.65%)	(4.79%)	(7.63%)	(13.72%)	(17.21%)	(2.76%)	(12.27%)	(31.72%)	(7.00%)
XAD	Advertising Expense	2.19%	0.00%	0.00%	0.00%	0.00%	0.00%	1.48%	0.00%	1.40%	0.53%	0.98%	0.71%
		(80.53%)	–	–	–	–	–	(78.02%)	–	(31.55%)	(2.23%)	(47.09%)	(218.28%)
XINT	Interest and Related Expense	0.44%	0.00%	1.79%	0.00%	1.57%	2.25%	0.00%	0.00%	0.00%	6.03%	0.00%	1.13%
		(97.21%)	–	(31.70%)	–	(35.48%)	(6.48%)	–	–	–	(12.67%)	–	(118.95%)
XRD	Research and Development Expense	0.44%	0.00%	1.79%	0.00%	1.57%	1.99%	0.00%	0.00%	0.00%	6.03%	0.00%	1.13%
		(97.21%)	–	(31.70%)	–	(35.48%)	(5.81%)	–	–	–	(12.67%)	–	(118.95%)
GP	Gross Profit (Loss)	63.60%	76.72%	72.70%	7.66%	60.32%	66.23%	8.89%	11.88%	45.56%	37.06%	2.93%	24.26%
		(7.55%)	(10.29%)	(10.77%)	(40.77%)	(8.47%)	(7.30%)	(13.25%)	(73.15%)	(5.59%)	(7.89%)	(24.57%)	(12.18%)
NI	Net Income (Loss)	0.88%	0.86%	0.26%	2.55%	1.57%	0.66%	2.24%	1.49%	0.70%	1.06%	1.87%	1.28%
		(1.01%)	(0.01%)	(100.00%)	(0.75%)	(7.56%)	(99.90%)	(1.60%)	(0.11%)	(2.05%)	(32.47%)	(9.25%)	(1.61%)
EPSPI	Earnings Per Share (Basic)	1.75%	0.86%	2.30%	2.98%	7.09%	2.78%	3.70%	0.50%	2.10%	5.14%	2.31%	4.54%
		(167.19%)	(34.78%)	(36.36%)	(82.80%)	(46.74%)	(89.69%)	(90.00%)	(747.37%)	(37.25%)	(200.00%)	(10.32%)	(47.61%)
EPSFI	Earnings Per Share (Diluted)	1.75%	0.86%	2.30%	2.98%	7.09%	2.78%	3.70%	0.50%	2.10%	5.14%	2.40%	4.54%
		(63.62%)	(14.49%)	(36.36%)	(82.80%)	(46.74%)	(89.69%)	(90.00%)	(747.37%)	(37.25%)	(200.00%)	(10.71%)	(47.61%)
OANCF	Operating Activities	2.19%	0.00%	4.34%	3.40%	3.94%	3.18%	8.21%	0.99%	3.51%	2.13%	2.40%	3.97%
		(0.57%)	–	(3.21%)	(1.34%)	(1.87%)	(1.65%)	(8.46%)	(1.41%)	(2.31%)	(10.15%)	(10.56%)	(7.08%)
FINCF	Financing Activities	0.44%	0.00%	1.02%	0.85%	0.79%	1.19%	5.19%	0.50%	0.47%	0.53%	0.89%	1.84%
		(0.01%)	–	(18.49%)	(128.36%)	(72.52%)	(1.57%)	(11.86%)	(1.15%)	(50.05%)	(34.01%)	(99.90%)	(10.99%)
IVNCF	Investing Activities	1.32%	0.00%	2.04%	3.40%	4.72%	1.85%	7.41%	0.99%	1.64%	1.60%	1.95%	2.70%
		(11.71%)	–	(167.78%)	(1.24%)	(23.28%)	(86.93%)	(17.63%)	(0.25%)	(33.51%)	(2.09%)	(11.50%)	(4.81%)
CHECH	Cash and Cash Equivalents - Increase (Decrease)	6.14%	2.59%	4.08%	1.70%	2.36%	2.91%	6.72%	0.00%	2.80%	1.24%	1.60%	3.55%
		(2.41%)	(23.27%)	(2.25%)	(1.17%)	(8.22%)	(1.51%)	(10.09%)	–	(5.36%)	(0.72%)	(3.30%)	(4.32%)

This table reports two number for each industry and variable. The first (upper) is the percentage of all discrepancy observations relative to all matched observations in the industry. The second (lower) is the median relative discrepancy value for the industry. For information about industry variables, please see Table 1.

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Table 5: Discrepancy statistics by industry and XBRL adoption phase

Industry	Phase I Filers					Phase II Filers					Phase III Filers				
		Discrepancies (as %)		Discrepancy Stats			Discrepancies (as %)		Discrepancy Stats			Discrepancies (as %)		Discrepancy Stats	
	Count	*All*	*Material*	*Mean*	*Median*	*Count*	*All*	*Material*	*Mean*	*Median*	*Count*	*All*	*Material*	*Mean*	*Median*
NoDur	445	13.26%	12.58%	18.03%***	6.72%	1,058	11.34%	10.87%	16.30%***	5.54%	3,979	11.69%	11.13%	49.33%***	6.01%
Durbl	150	16.00%	16.00%	36.06%***	12.19%	438	12.10%	12.10%	13.75%***	6.14%	2,260	12.30%	11.86%	26.50%***	6.57%
Manuf	1,006	12.43%	12.33%	58.67%***	11.00%	2,294	12.69%	12.60%	21.97%***	6.83%	6,524	12.58%	12.03%	32.98%***	6.71%
Enrgy	762	11.29%	10.37%	23.07%***	6.77%	607	10.87%	10.21%	26.33%***	9.79%	3,275	8.40%	7.79%	252.46%**	18.96%
Chems	333	12.91%	12.91%	16.06%***	8.03%	602	12.13%	11.63%	25.50%***	7.26%	2,209	12.18%	11.72%	450.36%**	7.01%
BusEq	1,254	12.28%	12.04%	20.75%***	6.92%	3,056	11.45%	10.93%	21.37%***	6.84%	14,451	11.41%	10.98%	31.07%***	6.52%
Telcm	402	11.19%	11.19%	24.05%***	8.46%	534	11.05%	9.74%	29.95%***	7.96%	2,082	9.37%	8.45%	103.01%***	5.67%
Utils	929	9.90%	9.36%	17.23%***	6.25%	1,307	12.62%	11.55%	27.29%***	7.41%	1,442	9.36%	8.32%	37.64%***	7.17%
Shops	837	9.32%	9.08%	12.98%***	3.39%	2,045	10.42%	9.49%	14.76%***	3.23%	6,995	10.32%	9.61%	336.48%**	4.90%
Hlth	837	11.35%	10.99%	29.24%**	7.15%	1,364	11.88%	10.70%	27.44%***	6.81%	10,884	9.09%	8.06%	26,696.64%*	7.35%
Money	1,246	14.93%	13.56%	26.31%***	9.41%	2,843	12.28%	11.26%	375.98%	7.11%	13,641	9.85%	8.67%	1,231.42%***	8.23%
Other	930	10.00%	9.46%	27.59%***	11.27%	2,735	8.85%	8.19%	24.49%***	5.30%	11,252	9.98%	9.11%	587.19%***	8.88%

Count – number of matched observations for specific industry and XBRL adoption phase.

All – percentage of all discrepancy observations for specific industry and phase.

Material – percentage of material discrepancy observations for specific industry and phase.

Mean, Median – mean and median statistics of relative value differences between Compustat and 10-K of discrepancy observations.

***, **, * indicate significance of mean difference t-tests at 99%, 95%, and 90% levels respectively.

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Table 6: Analysis of deviance of discrepancy observations

Panel A: Analysis of deviance with both Phase and Revenue variables included

Term	Observation level analysis						Company level analysis					
	All discrepancies			Material discrepancies			All discrepancies			Material discrepancies		
	χ^2	*df*	$\mathbb{P}(>\chi^2)$	χ^2	*df*	$\mathbb{P}(>\chi^2)$	χ^2	*df*	$\mathbb{P}(>\chi^2)$	χ^2	*df*	$\mathbb{P}(>\chi^2)$
Revenue	4.36	1	0.037*	6.52	1	0.011*	2.46	1	0.117	3.38	1	0.066+
Industry	62.94	11	0.000***	66.93	11	0.000***	65.67	11	0.000***	68.76	11	0.000***
Phase	1.06	2	0.588	0.98	2	0.61	0.95	2	0.623	0.88	2	0.646
Statement	945.61	2	0.000***	905.19	2	0.000***	717.66	2	0.000***	684.64	2	0.000***
Industry×Phase	44.40	22	0.003**	44.04	22	0.003**	46.14	22	0.002**	45.64	22	0.002**
Industry×Statement	445.08	22	0.000***	441.78	22	0.000***	318.87	22	0.000***	313.45	22	0.000***
Phase×Statement	13.54	4	0.009**	16.95	4	0.002**	9.54	4	0.049*	11.29	4	0.023*
Industry×Phase×Statement	78.61	44	0.001**	72.34	44	0.005**	70.96	44	0.006**	65.19	44	0.021*

Panel B: Analysis of deviance with Phase variable excluded

Term	Observation level analysis						Company level analysis					
	All discrepancies			Material discrepancies			All discrepancies			Material discrepancies		
	χ^2	*df*	$\mathbb{P}(>\chi^2)$	χ^2	*df*	$\mathbb{P}(>\chi^2)$	χ^2	*df*	$\mathbb{P}(>\chi^2)$	χ^2	*df*	$\mathbb{P}(>\chi^2)$
Revenue	10.87	1	0.001***	15.83	1	0.000***	6.94	1	0.008**	9.04	1	0.003**
Industry	87.28	11	0.000***	98.85	11	0.000***	95.89	11	0.000***	107.07	11	0.000***
Statement	1630.64	2	0.000***	1556.91	2	0.000***	1503.33	2	0.000***	1427.97	2	0.000***
Industry×Statement	834.72	22	0.000***	843.06	22	0.000***	634.29	22	0.000***	639.97	22	0.000***

***, **, *, + indicate significance of tests at 99.9%, 99%, 95%, and 90% levels respectively.

In all models, type III of sum of squares was utilized

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